

1. Social Media Analytics

Q.1 What is 'social media analytics' and how it is different from traditional Business analytics?

- ⇒
- Social media analytics is the process of extracting valuable insights from social media data to perform decision making.
 - By analyzing social media data, businesses can increase brand loyalty, generate leads, drive traffic and make forecasts.
 - It involves tracking metrics such as likes, shares, comments, mentions, sentiment and user demographics to make data-driven business analytics.

Social Media Analytics	Business Analytics
1) It involves the collection, analysis and interpretation of semi-structured and unstructured social media data.	1) Traditional business analytics uses structured and historical data.
2) Social media data is diverse, high-volume, real-time and third stored in third-party databases	2) Traditional business data is mostly stored in databases and spreadsheets in machine-readable format.
3) Public data and socialized in nature.	3) Private data and bureaucratic in nature.

Social Media data Analytics

Business Analytics

5) Data is widely shared over the internet.

5) Data is only shared within organization.

6) Boundary-less data (i.e. Boundary within the Internet).

6) Bound within the business intranet.

7) More sharing creates greater value/impact.

7) Less sharing creates more value.

8) No business control over data.

8) Tightly controlled by data business.

9) Data is high volume.

9) Data is medium to high volume.

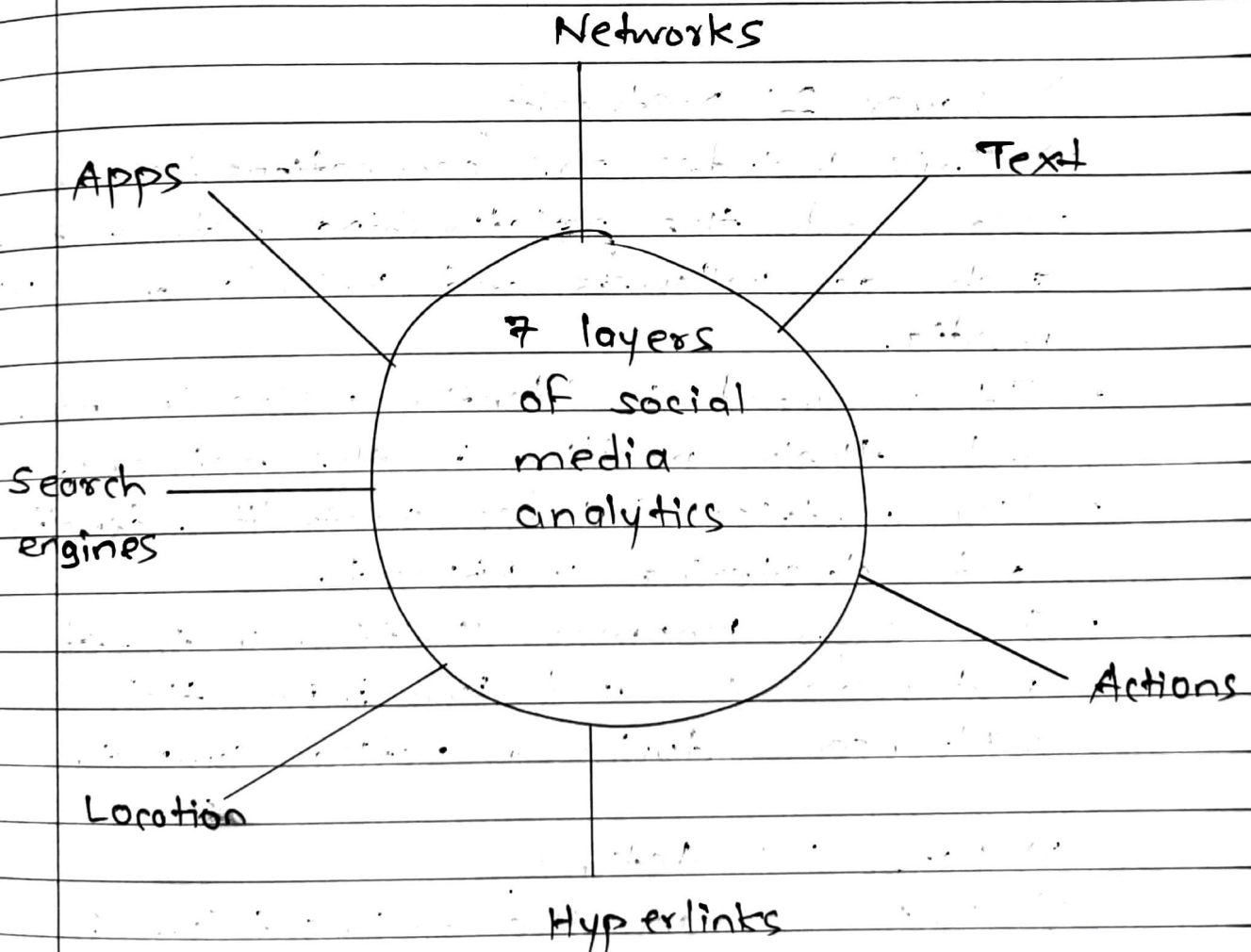
10) Data is informal in nature.

10) Data is Formal in nature.

Q.2 Explain briefly the seven layers of social media analytics with example.

⇒

- Social media has seven layers of data as shown below.
- Each level "contains potentially valuable information and insights that can be gathered ~~from~~ for business intelligence purposes.
- Out of the seven layers, some are visible or easily identifiable and other are invisible.



1) Layer 1: Text

⇒ This includes the content of social media posts, such as comments, tweets, blogs posts and Facebook status updates.

- It is used to understand user sentiment and identify emerging themes and topics.
- It involves NLP techniques to extract insights from written user-generated content.
- For e.g., Using sentiment analysis on tweets about an airline to determine customer satisfaction.

2) Layer 2: Networks

⇒ This includes the connections between users and the relationships between them, such as followers and friends on social media.

- It is used to identify influential nodes and their position in the network.
- Key metrics includes centrality, density and clustering coefficient.
- For e.g., Mapping Twitter retweet networks to find key influential influencers during a political campaign.

3) Layer 3: Actions

⇒ This includes the actions taken by users on social media, such as likes, comments, shares and other interactions.

- It is used to measure popularity, influence and prediction in social media.
- For e.g., Tracking likes and comments on an Instagram campaign to measure its popularity.

4) Layer 4: Mobile / Apps

- ⇒ Mobile analytics is the next frontier in the social business environment.
- This includes data related to the use of social media on mobile devices, such as the type of device used, the operating system and the location of the user.
- It is used to measure and optimize user engagement with mobile applications.
- For e.g., Instagram reels performance data accessed via Mobile vs Desktop.

5) Layer 5: Hyperlinks

- ⇒ This includes the links between social media posts and other websites, as well as the content of those linked websites.
- It is used to reveal Internet traffic patterns and sources of incoming or outgoing traffic.
- For e.g., Mapping how a blog post spreads through Facebook, Reddit and Twitter.

6) Layer 6: Location

- ⇒ This includes data on the geographical location of users and the location-specific content they engage with on social media.

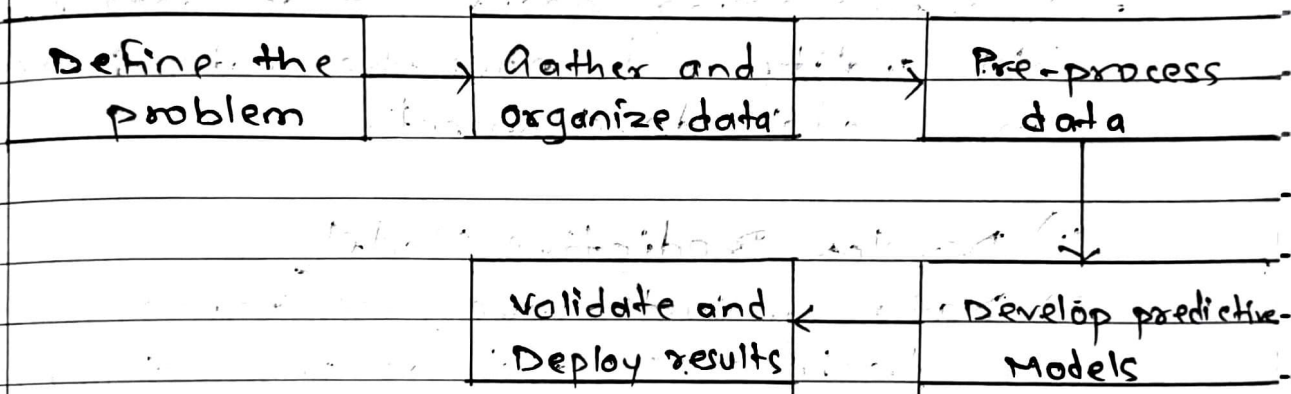
- It is used to mine and map the locations of social media users, content and data.
- For e.g., Analyzing tweets during natural disaster to locate affected areas in real-time.

7) Layer 7: Search: Engines

- ⇒ This includes data on how users find and access social media content through search engines, as well as the keywords they use in their searches.
- It is used to analyze historical search data for trends analysis, keyword monitoring and advertisement history.
- For e.g., Studying how a blog post shared on LinkedIn ranks on Google search.

Q.3 What is predictive analysis?

- Predictive analytics analyzes large amounts of aggregated social media data to predict future events.
- For example, analyzing social media data posts to predict future purchasing behavior or using historical website visits to predict future sales figure.
- How it works?



1) Define the problem

- ⇒ Firstly data analysts or data scientists define the problem.
- Defining the problem means clearly expressing the challenge that the organizing organization aims to focus using data analysis.

2) Gather and Organize data

- ⇒ Once you define a problem statement it is important to acquire and organize data properly.

- Acquiring data for predictive analytics means collecting and preparing relevant information and data from various sources like: databases, APIs, surveys and more that can be used to build and train models.

3) Pre-process data

- ⇒ Pre-processing involves cleaning the data to remove any kind of anomalies, handling missing data points and addressing outliers.
- Pre-processing ensures that data is of high quality and now the data is ready for model development.

4) Develop Predictive Model

- ⇒ Data analysts leverages range of tools or techniques such as regression models, decision trees to develop a predictive model.
- These models are trained on the prepared data to identify correlations and patterns.

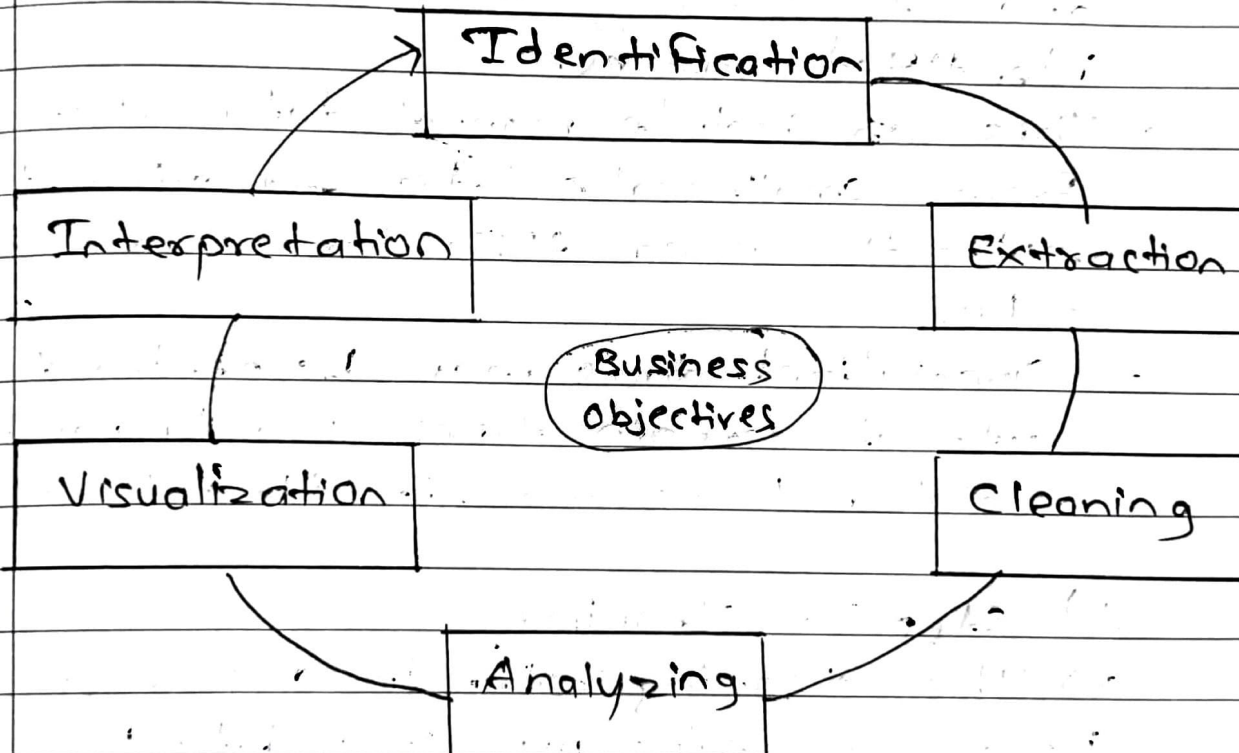
5) Validate and deploy results

- ⇒ After building the predictive model, the validation is critical steps to assess the accuracy and reliability of predictions.
- Evaluate model's performance on the test datasets.
- Once the model achieve satisfactory outcomes, it can be deployed to deliver predictions to stakeholders.

Q.4 Explain social media analytics cycle.

⇒

- The social media analytics process involves six steps to mine desired business insights from social media data.
- The process begins with defining business goals and objectives and continues until these objectives are fully satisfied.
- Social Media Analytical Cycle:



1) Step 1 - Identification

⇒ The Identification stage of social media involves finding the right sources of data to analyze in order to gain valuable business insights.

- The data should be aligned with business's objectives and can come from both official business-owned platforms, such as

- social media accounts and blogs, and non-official platforms such as Google search trends or Twitter search stream data.
- It is important to consider the business objectives when identifying the sources and types of data to be analyzed.

2) Step 2 - Extraction

- ⇒ The extraction step of social media analytics involves using appropriate methods and tools to gather data from identified sources.
- This can include manual data collection from small-scale data and automated extraction using APIs for larger data sets.
 - Specialized tools may be needed to extract certain types of data, such as social network and hyperlink network data.

3) Step 3 - Cleaning

- ⇒ The cleaning step in social media analytics involves removing unwanted data from the collected dataset.
- This can involve processes such as coding, filtering, clustering and NLP to remove irrelevant data.
 - Both automated and manual techniques may be used for cleaning.

4) Step 4 - Analyzing

- ⇒ The analyzing stage of social media analytics involves using clean data to identify valuable insights for the businesses.
- The approach and techniques used will depend on the type of data being analyzed and tools and algorithms employed.
- It is important to maintain integrity of the data while extracting meaningful insights.

5) Step 5 - Visualization

- ⇒ The visualization in social media analytics involves creating visual representation of the results of the analysis.
- Visualization can help reveal hidden patterns, relationships and trends in complex and large datasets.
- Types of visualization includes network data, trees, heatmaps, geospatial data.

6) Step 6 - Interpretation

- ⇒ Interpret and translate analytics results into a meaningful business problem.
- Two strategies or approaches used are:

a) Producing easily consumable analytical results.

b) Improving analytics consumption capabilities.

Q.6 Challenges of Social Media Analytics

⇒

- Social media data is in some ways a boon in terms of the amount, velocity, variety and insights it contains.
- However, their analysis and interpretation present some challenges.
- Analyzing unstructured data requires new metrics, tools and skills, especially for real-time analytics that most organizations lack.

1) Volume and Velocity

⇒ Social media data is big and generated quickly.

- Collecting and analyzing the millions of records that appear every second is extremely difficult.
- For example, Twitter sees 342,000 tweets every minute and Facebook shares 1 million likes every 20 minutes.
- Collecting all this information may not be practical.
- Knowing what to look for is important to narrow down the scope and size of your data.
- Fortunately, sophisticated tools have been developed to handle large amounts of fast data.

2) Diversity

- ⇒ Social media users and content they generate are highly diverse, multilingual and dependent on time and space.
- Not all tweets, likes or users are noteworthy.
 - Tweets and mentions from influential social media users are more important than those from non-influential users.
 - Due to the diverse nature of social media data, separating important content from noise can be difficult and time consuming.

3) Unstructuredness

- ⇒ Unlike data stored in corporate databases, which is mostly numeric, social media data is highly unstructured and consists of text, graphics, actions and relationships.
- Short social media texts, such as tweets and comments, have questionable grammatical structures and are cluttered with abbreviations, acronyms and emoticons, which can be extraction to represent intelligence has become a major challenge.

4) Data accuracy and reliability

- ⇒ Fake news, spams and bots distort social media data.
- Misinformation and biased opinions can lead to inaccurate conclusions.

- For e.g., a company analyzing customer sentiment may misinterpret bot-generated reviews and real feedback.

5) Real-time Data Processing

- ⇒ Social media data is constantly changing, requiring realtime analytics to stay relevant.
- Delayed analysis may lead to missed opportunities in marketing, crisis management or customer services.
- For e.g., A company failing to detect a trending negative hashtag about its brand may face PR damage.

6) High costs

- ⇒ Advanced analytical tools require high computational power and expertise.
- Small businesses may struggle to afford premium social media monitoring tools.
- For e.g., A startup lacks the budget to invest in AI-powered sentiment analysis tool.

Q.8 What ethical issues should be considered when mining social media data?

⇒

1) Privacy concerns

⇒ Many users share personal information on social media without expecting it to be analyzed.

- Collecting data without user consent can violate privacy rights.
- Every publicly available data might contain sensitive information.
- For e.g., A company scraping social media posts without informing users or obtaining permission.

2) Data security and protection

⇒ Collected data must be stored securely to prevent unauthorized access or data breaches.

- Businesses should comply with data protection laws. (e.g., GDPR, CCPA).
- For e.g., A social media analytics firm experiences a data leak, exposing users' private conversations.

3) Misuse of Data

⇒ Social media data can be used for unethical purposes like manipulation, surveillance or discrimination.

- Misinterpreting data trends can lead to biased conclusion.

- For e.g., A company unfairly targets job ads based on gender or age, leading to discrimination.

4) Social Impact

- ⇒ Excessive surveillance for analysis can lead to mental stress, self-censorship or reduced freedom of expression.
- AI-driven content targeting can create filter bubbles that reinforce biased viewpoints.
- For e.g., Social media platforms using algorithms that promote extreme content to increase engagement.

5) False News

- ⇒ Social media analytics can be misused to spread propaganda or misinformation.
- Companies must ensure they are not amplifying false narratives.
- For e.g., A political campaign uses social media data to spread misleading information to targeted groups.